

THE UNIVERSITY OF NEW SOUTH WALES
SCHOOL OF MATHEMATICS AND STATISTICS

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MATH3311/MATH5335

Mathematical Computing for Finance
Computational Methods for Finance

- (1) TIME ALLOWED – 2 HOURS
- (2) TOTAL NUMBER OF QUESTIONS – 4
- (3) ANSWER **QUESTIONS 1 AND 2.**
ANSWER **EITHER** QUESTION 3 **OR** QUESTION 4.
(ONLY ONE OF QUESTIONS 3 AND 4 CAN COUNT.)
- (4) THE QUESTIONS ARE OF EQUAL VALUE
- (5) THIS PAPER MAY BE RETAINED BY THE CANDIDATE
- (6) **ONLY** CALCULATORS WITH AN AFFIXED “UNSW APPROVED” STICKER
MAY BE USED

All answers must be written in ink. Except where they are expressly required pencils may only be used for drawing, sketching or graphical work.

1. [20 marks]

- i) Let
- C
- be an
- n
- by
- n
- covariance matrix, with ordered eigenvalues

$$\lambda_1 = 4.0 \cdot 10^{-4} \leq \lambda_2 = 4.8 \cdot 10^{-4} \leq \dots \leq \lambda_n = 3.6 \cdot 10^5.$$

- a) Is the matrix C positive definite?
- b) The MATLAB command `det(C)` yields 0. Is this correct? Provide an explanation for your answer and why $\det(C) = 0$.
- c) Find the condition number of the matrix C in an appropriate norm.
- d) Assume we want to solve the linear system $C\mathbf{x} = \mathbf{b}$ in double precision in MATLAB. The elements of C are known exactly, and the computed solution $\mathbf{x}$ must have at least 4 significant figures. How accurate must $\mathbf{b}$ be to achieve this?
- e) What are the largest and smallest singular values of C ?

- ii) Consider the matrix

$$B = I + \sum_{k=1}^K \alpha_k \mathbf{v}_k \mathbf{v}_k^\top,$$

where I is the $n \times n$ identity matrix, $K \ll n$, $\alpha_k \geq 0$, and $\mathbf{v}_k \in \mathbb{R}^n$ for $k = 1, \dots, K$.

- a) Show that B is positive definite.
- b) Why is explicitly forming B^{-1} not a good idea?
- c) How can you calculate $\mathbf{y} = B\mathbf{x}$ efficiently for a given $\mathbf{x} \in \mathbb{R}^n$.
- d) How many additions and multiplications are needed to compute $B\mathbf{x}$?

- iii) The present value at time
- t
- of \$1 received at time
- $T \geq t$
- is

$$P(t, T) = \exp\left(-\int_t^T r(s)ds\right) \quad \text{for } 0 \leq t \leq T. \quad (1)$$

i	1	2	3	4	5
t_i (years)	0	2	4	6	8
$r(t_i)$ (%)	3.9	3.6	3.4	3.8	4.1

Table 1: Interest rate data

- a) Use a quadrature rule of your choice and the data in Table 1 to estimate the present value at $t = 0$ of \$1 million received at $T = 8$.
- b) The Matlab command
`[z, w] = gauleg(N)`

calculates the N Gauss-Legendre nodes $z_j, j = 1, \dots, N$ and weights $w_j, j = 1, \dots, N$ for the interval $[-1, 1]$. Explain how these can be used to estimate $P(0, 8)$.

- c) What conditions must the integrand r satisfy for the Gauss-Legendre rule to work well when estimating $P(t, T)$?

2. [20 marks]

i) Consider the expression

$$\frac{f(a+h) - f(a)}{h} = f'(a) + O(h). \quad (2)$$

- a) Estimate the optimal stepsize h^* in terms of the relative machine precision ε when using the left-hand-side of (2) to estimate $f'(a)$.
- b) An analyst estimates the first derivative of a function using MATLAB at $a = 2$ million using $h = 10^{-12}$ and gets the value 0. Is this correct?

ii) We consider cubic splines.

- a) State the definition of a cubic spline $s : [a, b] \rightarrow \mathbb{R}$ with breakpoints $t_0, t_1, \dots, t_n$.
- b) Let the i th piece s_i of the cubic spline s be given by

$$s_i(t) = \frac{M_{i-1}}{6h_i}(t_i - t)^3 + \frac{M_i}{6h_i}(t - t_{i-1})^3 + \left(\frac{y_{i-1}}{h_i} - \frac{M_{i-1}h_i}{6} \right) (t_i - t) + \left(\frac{y_i}{h_i} - \frac{M_i h_i}{6} \right) (t - t_{i-1}), \quad (3)$$

where $h_i = t_i - t_{i-1}$. Show that $s(t_i) = y_i$ and $s''(t_i) = M_i$ for $i = 0, 1, \dots, n$.

- c) With s_i given by (3), derive a linear equation in the unknowns M_i such that $s'_i(t_i) = s'_{i+1}(t_i)$.
- d) Which additional conditions are satisfied for not-a-knot splines?
- e) Derive the linear system with unknowns $M_0, \dots, M_n$ which needs to be solved to define a not-a-knot spline.

iii) The standard normal cumulative distribution function (cdf) $\Phi : \mathbb{R} \rightarrow [0, 1]$ is

$$\Phi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x e^{-y^2/2} dy.$$

The inverse of the standard normal cdf is given by $\Phi^{-1} : [0, 1] \rightarrow \mathbb{R}$ such that $\Phi^{-1}(p) = x$, where $p = \Phi(x)$.

- a) For a given $p \in [0, 1]$, pose the problem of finding $\Phi^{-1}(p)$ as the solution of a nonlinear equation $f(x) = 0$.
- b) Calculate the derivatives $f'(x)$ and $f''(x)$.
- c) Prove that $f(x) = 0$ has a unique solution for any $0 < p < 1$.
- d) If $\bar{x}$ is an approximate value of the root x , with error $|x - \bar{x}| \approx 10^{-6}$, estimate the error after one iteration of the Newton method, which is quadratically convergent.

ANSWER EITHER QUESTION 3 OR QUESTION 4, NOT BOTH

3. [20 marks]

- i) Consider a real-valued random variable Y with a continuous, strictly increasing, cumulative distribution function F , so that

$$\mathbb{P}(Y \leq y) = F(y).$$

- a) Show that the random variable $X = F(Y)$ is uniformly distributed in the interval $[0, 1]$.
 b) Suppose we have a procedure to generate samples $x_1, x_2, \dots$ for X . How can we use these samples to generate samples from Y ? What is likely to be the main practical difficulty with this approach?
- ii) An asset price S_t is assumed to follow a geometric Brownian motion

$$\frac{dS_t}{S_t} = \mu dt + \sigma dW_t, \quad t \in (0, T], \quad (4)$$

where W_t is a Wiener process, μ, σ are positive constants and S_0 is given.

- a) What are the key properties of the Wiener process W_t ?
 b) Let the time interval be discretized by

$$t_i = i\Delta t, \quad i = 0, 1, \dots, m, \quad \Delta t = \frac{T}{m}.$$

Write down the formula for a discrete approximation $\tilde{S}_i \approx S_{t_i}$ for $i = 1, \dots, m$ of the geometric Brownian motion (4).

- c) Let $f(S_{t_0}, S_{t_1}, \dots, S_{t_m})$ be a given deterministic function of the asset prices at time steps $t_0, t_1, \dots, t_m$. Explain how you can estimate the expected value of f .
- iii) Consider a random vector $\mathbf{X} = (X_1, X_2, \dots, X_d)^\top \in \mathbb{R}^d$ with probability density function $p(\mathbf{x})$. Suppose we wish to compute a numerical approximation $E_N(f)$ to the expected value of some function $f(\mathbf{X})$,

$$E_N(f) \approx E(f) = \int_{\mathbb{R}^d} f(\mathbf{x})p(\mathbf{x}) d\mathbf{x}.$$

- a) How would you compute $E_N(f)$ using a Monte Carlo method with N samples.
 b) How might we get an idea of the accuracy of $E_N(f)$?
 c) Suppose we estimate that the error is about 10% using $N = 10^4$ points. How large a value of N should we expect to need if we want to reduce the error to about 1%?
 d) Explain the key ideas in using variance reduction with a function $g(\mathbf{x})$ to improve the accuracy of the estimate of $E_N(f)$.

ANSWER EITHER QUESTION 3 OR QUESTION 4, NOT BOTH

4. [20 marks] The value $V(S, t)$ of an option on an underlying asset with price S at time t satisfies the Black–Scholes PDE

$$\frac{\partial V}{\partial t} + rS \frac{\partial V}{\partial S} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} = rV \quad (5)$$

for $0 < S < \infty$ and $0 < t < T$. We assume that the prevailing risk-free interest rate r and the volatility σ are known positive constants.

Consider using (5) to price a put option, which gives the right to sell the underlying asset for the strike price K at expiry T .

- i) Carefully explain what the “initial” conditions are for this problem.
- ii) Carefully explain what the boundary conditions are for this problem.
- iii) Fix a maximum asset price $S_{\max} > 0$ and consider the domain $S \in [0, S_{\max}]$ and $t \in [0, T]$. Set up a grid $(S_j, t_\ell) = (j\Delta S, \ell\Delta t)$ for $j = 0, 1, 2, \dots, n+1$ and $\ell = 0, 1, 2, \dots, m$, where

$$\Delta S = \frac{S_{\max}}{n+1} \quad \text{and} \quad \Delta t = \frac{T}{m}.$$

Let $v_j^\ell \approx V(S_j, t_\ell)$ denote the approximate solution.

- a) At the time point t_ℓ and asset value S_j give
 - A) central difference approximations of $O(\Delta S^2)$ for the partial derivatives

$$\frac{\partial V}{\partial S} \quad \text{and} \quad \frac{\partial^2 V}{\partial S^2};$$

- B) a forward difference approximation of $O(\Delta t)$ for the partial derivative

$$\frac{\partial V}{\partial t}.$$

- b) We want to use an **implicit** finite difference method for computing an approximate solution v_j^ℓ .
 - A) Show that the finite difference equation has the form

$$\alpha_j v_{j-1}^\ell + \beta_j v_j^\ell + \gamma_j v_{j+1}^\ell = v_j^{\ell+1}, \quad (6)$$

and find the coefficients α_j , β_j and γ_j .

- B) What are the discrete “initial” conditions?
 - C) What are the discrete boundary conditions?

iv) After inserting the boundary data, Equation (6) has the form

$$A\mathbf{v}^\ell + \mathbf{b}_\ell = \mathbf{v}^{\ell+1}, \quad \text{where } \mathbf{v}^\ell = (v_1^\ell, \dots, v_n^\ell)^T.$$

- a) Define the entries $a_{i,j}$ of the matrix $A = (a_{i,j})$. What structure does the matrix A have?
- b) Is the matrix A sparse? Calculate its sparsity.
- c) Find the vector $\mathbf{b}_\ell$.
- d) How can the structure of the matrix A be exploited to make time-stepping as efficient as possible?
- e) Provide an estimation of the number of flops needed to compute the vector $\mathbf{v}^0$.

Q1)

i) a) Yes, all eigenvalues are > 0 .

b) No. $\det(C) = \lambda_1 \lambda_2 \dots \lambda_n$. Since $0 < \lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_n$, it follows that $\det(C) > 0$. However $\det(C)$ is so small such that it is represented as 0 in floating point representation.

c) Since C is symmetric and positive definite (as it is a covariance matrix), it follows that the 2-norm condition number is

$$k_2(C) = \frac{\lambda_n}{\lambda_1} = \frac{3.6e5}{4e-4} = 9e8.$$

d) The relative error in x is approximately $k_2(C) [\rho_C + \rho_b]$, where ρ_C is the relative error in C and ρ_b is the relative error in b . Since C is known 'exactly' we have $\rho_C \approx 2.2e-16$. We want 4 significant figures, hence the relative error in x should satisfy $\text{rel}(x) < \frac{1}{2} \cdot 10^{-4}$. Hence

$$k_2(C) [\rho_C + \rho_b] = 9 \cdot 10^8 [2.2 \cdot 10^{-16} + \rho_b] < \frac{1}{2} \cdot 10^{-4}$$

$$\Rightarrow \rho_b < \frac{1}{18} \cdot 10^{-12} - 2.2 \cdot 10^{-16} \approx$$

Thus if $\rho_b < \frac{1}{2} \cdot 10^{-13}$, i.e. b has 13 significant figures, then

$$\rho_b < \frac{1}{2} \cdot 10^{-13} < \frac{1}{18} \cdot 10^{-12} - 2.2 \cdot 10^{-16}.$$

e) Since C is symmetric and positive definite, the largest singular value is $3.6e5$ and the smallest singular value is $4e-4$ (same as largest and smallest eigenvalue).

(Q1) ii)

$$\begin{aligned} \textcircled{a}) \quad x^T B x &= x^T \cdot x + \sum_{k=1}^K \alpha_k (x^T v_k) \cdot (v_k^T x) \\ &= \|x\|_2^2 + \sum_{k=1}^K \alpha_k |x^T v_k|^2 \geq 0 \end{aligned}$$

for all $x \in \mathbb{R}^n \setminus \{0\}$, since $\alpha_k \geq 0$ and $\|x\|_2 > 0$. Thus B is

positive definite

b) Forming B^{-1} is likely to be a dense matrix, whereas B is very sparse.

c) We have $Bx = -x + \sum_{k=1}^K \alpha_k v_k (v_k^T x)$.

Computing $v_k^T x$ requires n multiplications and $n-1$ additions, for $k=1, \dots, K$.

Computing $\alpha_k (v_k^T x)$ requires $(n+1)$ multipl. and $n-1$ add.

Computing $\alpha_k (v_k^T x) v_k$ requires $2n+1$ multipl. and $n-1$ add.

Computing Bx requires

$K(2n+1)$ multiplications and $K(n-1+n)$ additions

Q1) iii)

a) We need to estimate $\int_0^8 r(s) ds$. Using a Trapezoidal Rule we get

$$\begin{aligned}\int_0^8 r(s) ds &\approx h \left[\frac{r_1}{2} + r_2 + r_3 + r_4 + \frac{r_5}{2} \right] \\ &= 2 \left[\frac{3.9}{2} + 3.6 + 3.4 + 4 + \frac{4.1}{2} \right] \% \\ &= 30\% = 0.3.\end{aligned}$$

Hence $P(0,8) \approx e^{-0.3}$

Using Simpson's method we get

$$\begin{aligned}\int_0^8 r(s) ds &\approx \frac{h}{3} [r_1 + 4r_2 + 2r_3 + 4r_4 + r_5] \\ &= \frac{2}{3} [3.9 + 4 \cdot 3.6 + 2 \cdot 3.4 + 4 \cdot 4 + 4.1] \% \\ &= \frac{2}{3} \cdot 45.2 \% \\ &= 0.3013\end{aligned}$$

Hence $P(0,8) \approx e^{-0.3013}$

b) We use a linear transformation $s(z) = \alpha + \beta z$, such that $s(-1) = 0$, and $s(1) = 8$. Thus $\alpha - \beta = 0 \Leftrightarrow \alpha = \beta$ and $\alpha + \beta = 8$.

Hence $\alpha = \beta = 4$ and therefore $s(z) = 4 + 4z$ and $\frac{ds}{dz} = 4$. Thus

$$\int_0^8 r(s) ds \approx 4 \sum_{j=1}^N w_j r(4 + 4z_j).$$

c) The Gauss-Legendre rule with N points has degree of precision $2N-1$ if the integrand has $2N$ continuous derivatives on the interval of integration. So the integrand should have 'many' derivatives.

Q2) i)

a) We have approximation error $O(h)$ and rounding error $O(\frac{\epsilon}{h})$.

Hence $E(h) = c_1 h + c_2 \frac{\epsilon}{h}$.

Minimizing yields $h = O(\epsilon^{1/2})$.

$$E'(h) = c_1 - c_2 \frac{\epsilon}{h^2} = 0$$

$$\Rightarrow h = \left(\frac{c_2 \epsilon}{c_1}\right)^{1/2} = O(\epsilon^{1/2}).$$

Minimum, as $E''(h) = -c_2 \frac{\epsilon}{h^3} < 0$.

b) Since $a = 2e6$ and $h = 1e-18$, we get $a+h = 2e6 + 1e-18 = (2e0 + 1e-18)10^6$.

However, in double precision the machine epsilon is $\approx 2.2e-16$. Hence in floating point representation $a+h = a$ and so $f(a+h) - f(a) = 0$.

The result 0 is therefore meaningless.

ii) A function $s: [a, b] \rightarrow \mathbb{R}$ is a spline of order 4 (cubic spline)

if there exist polynomials $s_1, \dots, s_n$, each of degree at most 3, such that

1) $s(t) = s_i(t)$ for $t_{i-1} \leq t \leq t_i$ for $i=1, 2, \dots, n$;

2) s and its derivatives s', s'' exist and are continuous on $[a, b]$.

b) We have

$$s_i(t_i) = \frac{M_i}{6h_i} (t_i - t_{i-1})^3 + \left(\frac{y_i}{h_i} - \frac{M_i h_i}{6}\right) (t_i - t_{i-1}) = y_i,$$

$$s_i(t_{i-1}) = \frac{M_{i-1}}{6h_i} (t_i - t_{i-1})^3 + \left(\frac{y_{i-1}}{h_i} - \frac{M_{i-1} h_i}{6}\right) (t_i - t_{i-1}) = y_{i-1},$$

hence $s_{i-1}(t_{i-1}) = s_i(t_{i-1}) = y_{i-1}$ and hence $s(t_{i-1}) = y_{i-1}$ for all i .

We have

$$s_i''(t) = \frac{M_{i-1}}{h_i}(t_i - t) + \frac{M_i}{h_i}(t - t_{i-1}).$$

Hence $s_i''(t_i) = M_i$ and $s_i''(t_{i-1}) = M_{i-1}$. Thus $s''(t_i) = M_i$.

$$c) s_i'(t) = -\frac{M_{i-1}}{2h_i}(t_i - t)^2 + \frac{M_i}{2h_i}(t - t_{i-1})^2 - \frac{y_{i-1}}{h_i} + \frac{M_{i-1}h_i}{6} + \frac{y_i}{h_i} - \frac{M_ih_i}{6}$$

$$s_i'(t_i) = \frac{M_i}{2}h_i + \frac{M_{i-1}h_i}{6} - \frac{M_ih_i}{6} + \frac{y_i - y_{i-1}}{h_i} = \frac{M_ih_i}{3} + \frac{M_{i-1}h_i}{6} + \frac{y_i - y_{i-1}}{h_i}$$

$$s_i'(t_{i-1}) = -\frac{M_{i-1}h_i}{2} + \frac{M_{i-1}h_i}{6} - \frac{M_ih_i}{6} + \frac{y_i - y_{i-1}}{h_i} = -\frac{M_{i-1}h_i}{3} - \frac{M_ih_i}{6} + \frac{y_i - y_{i-1}}{h_i}$$

Thus $s_i'(t_i) = s_{i+1}'(t_i)$ if and only if

$$\frac{M_ih_i}{3} + \frac{M_{i-1}h_i}{6} + \frac{y_i - y_{i-1}}{h_i} = -\frac{M_{i+1}h_{i+1}}{3} - \frac{M_ih_{i+1}}{6} + \frac{y_{i+1} - y_i}{h_{i+1}}$$

$$\frac{h_i + h_{i+1}}{3}M_i + \frac{h_i}{6}M_{i-1} + M_{i+1}\frac{h_{i+1}}{6} = \frac{y_{i+1} - y_i}{h_{i+1}} - \frac{y_i - y_{i-1}}{h_i}$$

$$d) s_1'''(t_1) = s_2'''(t_1) \text{ and } s_{n-1}'''(t_{n-1}) = s_n'''(t_{n-1})$$

$$e) s_i'''(t) = -\frac{M_{i-1}}{h_i} + \frac{M_i}{h_i}. \text{ Thus } s_1'''(t_1) = s_2'''(t_1) \text{ is equivalent to}$$

$$-\frac{M_0}{h_1} + \frac{M_1}{h_1} = -\frac{M_1}{h_2} + \frac{M_2}{h_2} \text{ and } s_{n-1}'''(t_{n-1}) = s_n'''(t_{n-1}) \text{ is equivalent to}$$

$$-\frac{M_{n-2}}{h_{n-1}} + \frac{M_{n-1}}{h_{n-1}} = -\frac{M_{n-1}}{h_n} + \frac{M_n}{h_n}.$$

Hence the linear system is

$$-\frac{1}{h_1} M_0 + \left(\frac{1}{h_1} + \frac{1}{h_2}\right) M_1 - \frac{1}{h_2} M_2 = 0$$

$$\frac{h_i}{6} M_{i-1} + \frac{h_i + h_{i+1}}{3} M_i + \frac{h_{i+1}}{6} M_{i+1} = \frac{y_{i+1} - y_i}{h_{i+1}} - \frac{y_i - y_{i-1}}{h_i} \quad \text{for } i=0, \dots, 1$$

$$-\frac{1}{h_{n-1}} M_{n-2} + \left(\frac{1}{h_{n-1}} + \frac{1}{h_n}\right) M_{n-1} - \frac{1}{h_n} M_n = 0.$$

$$\text{iii)} \quad \Phi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x e^{-\frac{y^2}{2}} dy$$

a) For a given $p \in [0, 1]$, we want to find x such that $\Phi^{-1}(p) = x$, which is the same as $p = \Phi(x)$. Let

$$f(x) = \Phi(x) - p.$$

Then $f(x=0) (=) \Phi(x) = p (=) \Phi^{-1}(p) = x$.

b) We have $f'(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}$ and $f''(x) = -\frac{x}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}$.

c) We have $f'(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} > 0 \quad \forall x \in \mathbb{R}$. Hence, $f'(x) \neq 0$.

Thus f is monotonically strictly increasing. Further $\lim_{x \rightarrow -\infty} f(x) = -p < 0$ and $\lim_{x \rightarrow \infty} f(x) = 1-p > 0$. Thus $f(x) = 0$ has a unique solution.

d) After one iteration, the error is of order $(10^{-6})^2 = 10^{-12}$.

Q3) i)

a) We have

$$\mathbb{P}(X \leq x) = \mathbb{P}(F(Y) \leq x) = \mathbb{P}(Y \leq F^{-1}(x)) = F(F^{-1}(x)) = x, \quad x \in [0, 1]$$

Hence X is uniformly distributed.

b) The samples $y_1 = F^{-1}(x_1), y_2 = F^{-1}(x_2), \dots$ are samples from Y .

The function F^{-1} is often difficult to compute numerically since in many cases there is no explicit formula for it.

ii) a) $\rightarrow W_t$ depends continuously on t

$\rightarrow W_0 = W_0 = 0$ (with probability 1)

$\rightarrow W_t - W_s \sim N(0, t-s)$ for $0 \leq s < t \leq T$.

$\rightarrow W_t - W_s$ and $W_v - W_u$ are independent for $0 \leq s < t < u < v \leq T$.

b) We have

$$\cancel{\$t_i} \quad \cancel{\$t_{i+1}} \quad \frac{\Delta S_{t_i}}{S_{t_i}} = \mu \Delta t + \sigma \Delta W_t = \mu \Delta t + \sigma \sqrt{\Delta t} Z_i,$$

where $Z_i \sim N(0, 1)$ and $\Delta S_{t_i} = S_{t_{i+1}} - S_{t_i}$. Hence

$$(*) \quad S_{t_{i+1}} = S_{t_i} [1 + \mu \Delta t + \sigma \sqrt{\Delta t} Z_i], \quad \text{for } i = 0, 1, \dots, m-1.$$

c) For $j = 1, \dots, N$ we generate asset paths

$$S_{t_i}^{(j)}, \quad i = 0, \dots, m-1 \quad \text{using } (*): \quad S_{t_{i+1}}^{(j)} = S_{t_i}^{(j)} [1 + \mu \Delta t + \sigma \sqrt{\Delta t} Z_i^{(j)}]$$

where the $Z_i^{(j)}$ are chosen independently. Then compute

$\frac{1}{N} \sum_{j=1}^N \phi(S_0^{(j)}, \dots, S_m^{(j)})$, where ϕ is now a function of the discretized asset prices. Now we increase N to get a better and better approximation.

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ii) a) Generate samples $x_1, \dots, x_N$ ^{i.i.d.} with pdf $p(x)$. Then estimate

$$E(f) = \int_{\mathbb{R}^S} f(x) p(x) dx \approx E_N(f) = \frac{1}{N} \sum_{n=1}^N f(x_n).$$

b) The estimator $E_N(f)$ is unbiased. To estimate $E(E_N(f) - E(f))^2$, compute the sample variance of $f(x_1), \dots, f(x_N)$.

c) The convergence of Monte Carlo is of order $N^{-\frac{1}{2}}$. So if $0.1 \leftarrow N^{-\frac{1}{2}}$, then $0.01 = c N^{-\frac{1}{2}}$, then $0.01 = c 10^{-1} N^{-\frac{1}{2}} = c (10^2 N)^{-\frac{1}{2}}$.

Thus $N \approx 10^6$.

d) We choose a function $g(x)$ which is close to $f(x)$ and for which $\int_{\mathbb{R}^S} g(x) p(x) dx$ can be computed explicitly. Then we estimate

$$\int_{\mathbb{R}^S} f(x) p(x) dx \approx \int_{\mathbb{R}^S} g(x) p(x) dx + \frac{1}{N} \sum_{n=1}^N (f(x_n) - g(x_n)).$$

Q4)

i) $V(S, T) = \max(S - K, 0)$

ii) $V(0, t) = 0$

iii) $V(S, t) = S - Ke^{-r(T-t)}$ as $S \rightarrow \infty$.

iii) a)

A) $\frac{\partial V}{\partial S} : \frac{v_{j+1}^l - v_j^l}{2 \Delta S} ; \quad \frac{\partial^2 V}{\partial S^2} : \frac{v_{j+1}^l - 2v_j^l + v_{j-1}^l}{\Delta S^2}$

B) $\frac{\partial V}{\partial t} : \frac{v_j^{l+1} - v_j^l}{\Delta t}$

b) A) We have

$$\frac{v_j^{l+1} - v_j^l}{\Delta t} + r S_j \frac{v_{j+1}^l - v_j^l}{2 \Delta S} + \frac{1}{2} \sigma^2 S_j^2 \frac{v_{j+1}^l - 2v_j^l + v_{j-1}^l}{\Delta S^2} = r v_j^l$$

By rearranging we get

$$v_j^{l+1} = \left(\frac{r S_j \Delta t}{2 \Delta S} - \frac{\Delta t \sigma^2 S_j^2}{2 \Delta S^2} \right) v_{j-1}^l + \left(1 + r \Delta t + \frac{\Delta t \sigma^2 S_j^2}{\Delta S^2} \right) v_j^l + \left(-\frac{\Delta t r S_j}{2 \Delta S} - \frac{\Delta t \sigma^2 S_j^2}{2 \Delta S^2} \right) v_{j+1}^l$$

Therefore

$$\alpha_j = \frac{r S_j \Delta t}{2 \Delta S} - \frac{\Delta t \sigma^2 S_j^2}{2 \Delta S^2}, \quad \beta_j = 1 + r \Delta t + \frac{\Delta t \sigma^2 S_j^2}{\Delta S^2},$$

$$\gamma_j = -\frac{\Delta t r S_j}{2 \Delta S} - \frac{\Delta t \sigma^2 S_j^2}{2 \Delta S^2};$$

$$B) \quad v_j^m = \max(S_j - K, 0)$$

$$C) \quad v_0^l = 0$$

$$v_{n+1}^l = S_{\max} - K e^{-r(T-t_l)}$$

iv) a) The matrix A is an $n \times n$ matrix with

$$a_{j,j-1} = \alpha_j, \quad j=2,3,\dots,n$$

$$a_{j,j} = \beta_j, \quad j=1,2,\dots,n$$

$$a_{j,j+1} = \gamma_j, \quad j=1,2,\dots,n-1;$$

The remaining elements are 0. Hence the matrix is tri-diagonal.

b) Yes, the matrix A is sparse, with sparsity

$$\frac{n^2 - (n + 2(n-1))}{n^2} = 1 - \frac{3}{n} + \frac{2}{n^2}.$$

c) The vector $\underline{b}$ is

$$\underline{b} = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ S_{\max} - K e^{-r(T-t_l)} \end{pmatrix}$$

d) At each time step we need to solve $A v^l = v^{l+1} - \underline{b}$. Since A is tri-diagonal, we can use the Thomas algorithm, which uses $\mathcal{O}(n)$ operations.

e) Solving the linear system requires $\mathcal{O}(n)$ operations. Since we solve this system $\mathcal{O}(m)$ times, we need $\mathcal{O}(mn)$ operations.